

JKBOSE 2026

CLASS 11TH

THE LIVING WORLD

LECTURE- 03

BOTANY

ARSHID BHAT SIR



Topics To Covered

- 1** Biodiversity ✓
- 2** Need of Classification ✓
- 3** Taxonomy ✓
- 4** Systematics ✓
- 5** Nomenclature ✓

↓
Among plants, Angiosperms (Flowering plants)
are maximum.

NEED OF CLASSIFICATION

- ↳ There is huge number of plant and animal species which vary from each other.
- ↳ It becomes very difficult to study individually all of them.
- ↳ In order to make their study easy — organisms have been classified into various groups/categories on the basis of easily observable characters.

Q. What is classification?

Ans. To organise organisms into different groups.

TAXONOMY

- ↳ Taxonomy — derived from two Greek words $\left\{ \begin{array}{l} \text{"Taxis"} \text{ — Arrangement} \\ \text{"nomos"} \text{ — laws/rules.} \end{array} \right.$
- ↳ Taxonomy — is a branch of science which deals with Classification of living organisms by following certain rules/laws.
- ↳ Carolus Linnaeus — is regarded as father of taxonomy.
- ↳ Term "taxonomy" — Coined by "A.P deCandolle" in his book $\downarrow$ "Theory of Elementary botany"

Taxonomy — Two types:

a. Classical taxonomy :-

= involves use of morphological (ie external) characters for classification of living organisms.

b. Modern taxonomy :-

= involves use of :

- i) Morphological characters (external characters)
- ii) Anatomical " (internal ")
- iii) Cytological " (cell ")
- iv) Physiological "
- v) Developmental "
- vi) Ecological characters etc.

STEPS INVOLVED IN TAXONOMY

↳ 04 Steps:

- a. Characterisation.
- b. identification.
- c. Classification.
- d. Nomenclature

Note:

Taxonomy = Characterisation + identification + Classification + Nomenclature

a. Characterisation :-

Here organism is described according to its all morphological, Cytological, developmental, behavioural etc. properties

Account OR of characters.

b. Identification :-

process by which an organism is recognised from already known organisms

Comparison OR with other known species.

c. Classification :-

Process by which an organism is grouped into convenient categories on the basis of some easily observable characters
OR.
 Assign a group to an organism.

d. Nomenclature :-

It is the naming of an organism according to certain established rules. (if the sps. is newly discovered).

NEET

Q. Practical purpose of taxonomy / classification?

Ans. To facilitate the identification of unknown organisms.

SYSTEMATICS

* Systematics — derived from latin word — "Systema"
 ↓ means

≡ Term "Systematics" — Coined by "Carolous Linneous" in his book
 ↓
 "Systema naturae"

* Systematics deals with $\left\{ \begin{array}{l} \text{diversity of living organisms} \\ \text{AND} \\ \text{Evolutionary relationship b/w living organisms.} \end{array} \right.$

* Modern Systematics involves Taxonomy AND Phylogeny (evolutionary relationship among organisms).

NOTE:

$$\text{Systematics} = \text{Taxonomy} + \text{Phylogeny}$$

NOMENCLATURE



↳ process of assigning a name to an organism.

↳ Two types of name:

a. Common name / Vernacular name / local names :

↳ Names given to organisms in a particular language or region.

↳ These names vary from region to region or from language to language and thus creates confusion.

b. Biological name / Scientific name :-

↳ Names given to organisms by taxonomists

↳ Same throughout world

↳ No confusion (ambiguity)

BINOMIAL NOMENCLATURE

↳ given by "Carolus Linneus" in his book "Species Plantarum"
↳ As per binomial nomenclature — the scientific name of an organism possess two components:

a. Generic name. ✓

b. Specific epithet. ✓

Ex. Potato — Common name.
↓ scientific name

Solanum tuberosum.

RULES OF BINOMIAL NOMENCLATURE

- ①. Biological / scientific names are taken from latin language (or are latinised).
- ②. Biological names consists of two components $\left\{ \begin{array}{l} \text{Generic name} \\ \text{Specific epithet} \end{array} \right.$

Ex. Solanum tuberosum. (potato)

↓ ↓

Generic name Specific epithet.

③. First letter of Generic name — Capital.

First letter of AND specific epithet — Small.

Ex. Brinjal

Solanum melongena

①. Scientific names when handwritten — Should be underlined separately

This indicates the latin origin of scientific name

Scientific names AND should be typed in italics

⑤. Name of Author / discoverer → is written after specific epithet.
 → neither underlined nor italicized.
 → written in abbreviated form.

Ex. "Mango" — Mangifera indica Linn. → Linneous
 "Author / discoverer".

"Human" — Homo sapiens Linn.

Note :-

- ① ICZN = "International Code of Zoological Nomenclature" (Animals)
- ② ICBN = "International Code for Botanical Nomenclature" (plants)

QUESTIONS

Who wrote systema naturae:

- ☒ 1 Linnaeus
- ☐ 2 Mayr
- ☐ 3 John Ray
- ☐ 4 De Candolle

QUESTIONS

'X' and 'Y' are the components of Binomial nomenclature. This naming system was proposed by 'Z': [2023 Manipur]

- ✓ **1** X-Generic name, Y-Specific epithet, Z-Carolus Linnaeus
- 2** X-Specific epithet, Y-Generic name, Z-R.H. Whittaker
- 3** X-Specific epithet, Y-Generic name, Z-Carolus Linnaeus
- 4** X-Generic name, Y-Specific epithet, Z-R.H. Whittaker

QUESTIONS

Select the correctly written scientific name of Mango which was first described by Carolus Linnaeus. [MR* 2019]

1 *Mangifera indica* Car. Linn.

✓ 2 *Mangifera indica* Linn.

3 *Mangifera indica*

4 *Mangifera Indica*

QUESTIONS

Biodiversity of a geographical region represents:

[2011 Mains]

- 1 Genetic diversity present in the dominant species of the region ✕
- 2 Species endemic to the region ✕
- 3 Endangered species found in the region ✕
- 4 ✓ The diversity in the organisms living in the region

QUESTIONS

Phylogeny refers to:

- 1 Natural classification ✗
- 2 Evolutionary classification ✗
- 3 Evolutionary history ✓
- 4 Origin of algae

Thank
You